

Lab 5
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6B1

```
[num, den] = series (num1, den1, num2, den2)
[num, den] = parallel (num1, den1, num2, den2)
[num, den] = feedback (num1, den1, num2, den2)
[num, den] = cloop (num1, den1, -1)
```

Ex 1

a)

```
untitled2.m  x  +
1
2      [num, den] = series ([10],[1 2 10],[5], [0 1 5])
3      [num, den] = parallel ([10],[1 2 10],[5], [0 1 5])
4      [num, den] = feedback ([10],[1 2 10],[5], [0 1 5])
5      [num, den] = cloop ([10],[1 2 10],-1)
```

Command Window

```
Warning: "C:\Users\hp\OneDrive\Desktop\ADC Project" selected using the Last Wo
Using "C:\Users\hp\OneDrive\Documents\MATLAB" as the initial working folder in
>> untitled2
```

```
num =

    0    0    0    0    50

den =

    0    1    7    20    50
```

b,c,d

```
num =

    0    0    5    20    100

den =

    0    1    7    20    50

num =

    0    0    0    10    50

den =

    0    1    7    20    100

num =

    0    0    10

den =

    1    2    20
```

Day: _____

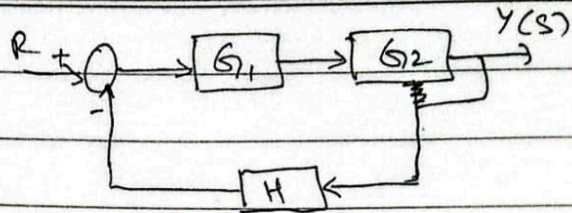
Date: _____

* Ex:-2

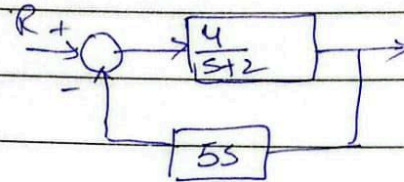
$$G_1(s) = 4$$

$$G_2(s) = \frac{1}{s+2}$$

$$H(s) = 5s$$



$$T(s) = \frac{\frac{4}{s+2}}{1 + \frac{4}{s+2} (5s)}$$



$$= \frac{\frac{4}{s+2}}{s+2 + 20s} = \frac{4}{s+2+10s} = \frac{4}{21s+2}$$

Series $\rightarrow \frac{4}{s+2}$

↓
feedback

Ex:-1

a) $\frac{50}{s^2 + 7s + 20s + 50}$

b) $\frac{5s^2 + 20s + 100}{s^3 + 7s^2 + 20s + 50}$

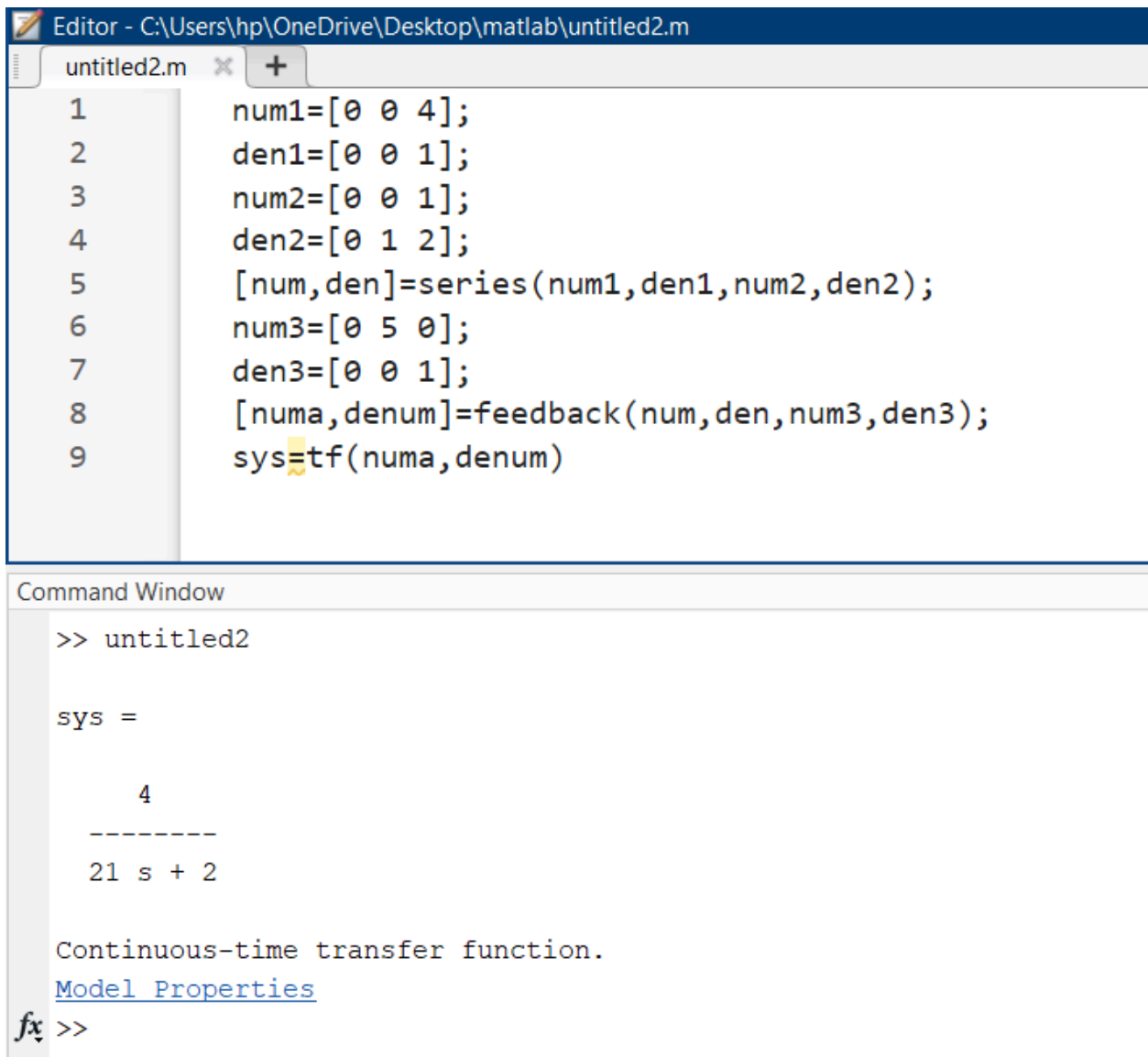
c) $\frac{10s + 50}{s^3 + 7s^2 + 20s + 100}$

d) $\frac{10}{s^2 + 2s + 20}$

Ex2

```
9      % Series connection G1*G2
10     [num1, den1] = series([4],[1], [1], [0 1 2]);
11
12     % Feedback system
13     [num2, den2] = feedback([0 0 4],[0 1 2], [0 5 0],[0 0 1]);
14
```

In detail ex2



The image shows a MATLAB Editor window titled "Editor - C:\Users\hp\OneDrive\Desktop\matlab\untitled2.m" with a tab for "untitled2.m". The code in the editor is as follows:

```
1      num1=[0 0 4];
2      den1=[0 0 1];
3      num2=[0 0 1];
4      den2=[0 1 2];
5      [num,den]=series(num1,den1,num2,den2);
6      num3=[0 5 0];
7      den3=[0 0 1];
8      [numa,denum]=feedback(num,den,num3,den3);
9      sys=tf(numa,denum)
```

Below the editor is the Command Window, which shows the execution of the script "untitled2". The output is:

```
>> untitled2

sys =

      4
-----
21 s + 2

Continuous-time transfer function.
Model Properties
fx >>
```

Ex3

```
Editor - C:\Users\hp\OneDrive\Desktop\matlab\untitled2.m
untitled2.m x +
1 num1=[0 1 0];
2 den1=[0 0 1];
3 num2=[0 1 0];
4 den2=[0 0 1];
5 [num3,den3]=series(num1,den1,num2,den2);
6 sys1=tf(num3,den3)
7 num4=[0 0 1];
8 den4=[0 1 0];
9 [num5,den5]=parallel(num3,den3,num4,den4);
10 sys2=tf(num5,den5)
11 [num6,den6]=cloop(num5,den5,-1);
12 sys3=tf(num6,den6)
13 num7=[0 0 1];
14 den7=[0 1 0];
15 [num8,den8]=series(num6,den6,num7,den7);
16 sys4=tf(num8,den8)
17 num9=[0 1 0];
18 den9=[0 0 1];
19 [num,den]=feedback(num8,den8,num9,den9);
20 sys=tf(num,den)
```

sys1-> series
sys2-> parallel
Sys3 ->cloop
sys4->series
sys->feedback

Output:

```
sys1 =
```

$$s^2$$

Continuous-time transfer function.

[Model Properties](#)

```
sys2 =
```

$$\frac{s^3 + 1}{s}$$

Continuous-time transfer function.

[Model Properties](#)

```
sys3 =
```

$$\frac{s^3 + 1}{s^3 + s + 1}$$

fx

```
sys4 =
```

$$\frac{s^3 + 1}{s^4 + s^2 + s}$$

Continuous-time transfer function.

[Model Properties](#)

```
sys =
```

$$\frac{s^3 + 1}{2 s^4 + s^2 + 2 s}$$

Continuous-time transfer function.

[Model Properties](#)

⌘ ↵

Ex4

```
Editor - C:\Users\hp\OneDrive\Desktop\matlab\untitled2.m
untitled2.m x +
1      %(a)
2      num=[0 1 0];
3      den=[1 14 56 160];
4      [A,B,C,D]=tf2ss(num,den)
5      %(b)
6      num=[1 7 2];
7      den=[1 9 26 24];
8      [A,B,C,D]=tf2ss(num,den)
```

a)

```
Command Window
A =
    -14    -56   -160
     1      0      0
     0      1      0

B =
     1
     0
     0

C =
     0      1      0

fx
```

A remaining and **B) output**

```
D =
     0

A =
    -9    -26   -24
     1      0      0
     0      1      0

B =
     1
     0
     0

C =
     1      7      2

D =
     0
```

Ex5

a)

```
untitled2.m x +
1 A=[0 1;-6 -5];
2 B=[0;1];
3 C=[8 1];
4 D=[0];
5 [num,den]=ss2tf(A,B,C,D)
6 sys=tf(num,den)

Command Window

den =

    1.0000    5.0000    6.0000

sys =

      s + 8
-----
s^2 + 5 s + 6
```

B)

```
untitled2.m x +
1 A=[0 1 0;0 0 1;-1 -2 -3];
2 B=[10;0;0];
3 C=[1 0 0];
4 D=[0];
5 [num,den]=ss2tf(A,B,C,D)
6 sys=tf(num,den)

Command Window

den =

    1.0000    3.0000    2.0000    1.0000

sys =

    10 s^2 + 30 s + 20
-----
s^3 + 3 s^2 + 2 s + 1

Continuous-time transfer function.
Model Properties
fx >>
```

Ex 6

Editor - C:\Users\hp\OneDrive\Desktop\matlab\untitled2.m
untitled2.m

```

1      num1=[0 5 7];
2      den1=[0 1 7];
3      num2=[0 0 1];
4      den2=[1 5 4 0];
5      [num3,den3]=series(num1,den1,num2,den2);
6      sys1=tf(num3,den3)
7      num4=[0 0 10];
8      den4=[0 0 1];
9      [num,den]=feedback(num3,den3,num4,den4);
10     sys=tf(num,den)
11     [A,B,C,D]=tf2ss(num,den)

```

Command Window

```

>> untitled2

sys1 =

          5 s + 7
-----
s^4 + 12 s^3 + 39 s^2 + 28 s

Continuous-time transfer function.
Model Properties

sys =

          5 s + 7
-----

```

sys1-> series
Sys -> feedback

sys1 =

$$\frac{5s + 7}{s^4 + 12s^3 + 39s^2 + 28s}$$

Continuous-time transfer function.

[Model Properties](#)

sys =

$$\frac{5s + 7}{s^4 + 12s^3 + 39s^2 + 78s + 70}$$

Continuous-time transfer function.

[Model Properties](#)

A =

$$\begin{bmatrix} -12 & -39 & -78 & -70 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

B =

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

C =

$$\begin{bmatrix} 0 & 0 & 5 & 7 \end{bmatrix}$$

D =

$$\begin{bmatrix} 0 \end{bmatrix}$$